

Smart Lender – AI-Powered Loan Approval Prediction System

Final Project Report

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	

- GitHub Repository: https://github.com/anushaobulapuram/smart_lender_smart_bridge
- Project Drive Link:
<https://drive.google.com/file/d/1SFsukLL6qTV93eniIDsgYuXs0K5jTuJl/view?usp=drivesdk>

1. Introduction

Project Overview: Smart Lender AI is a Machine Learning-based web application that predicts whether a loan application should be approved or rejected, based on applicant details such as income, credit history, education and property area.

Purpose: To help banks and financial institutions automate the loan approval process, reduce manual effort, improve decision-making accuracy, and provide a faster, more transparent evaluation for applicants.

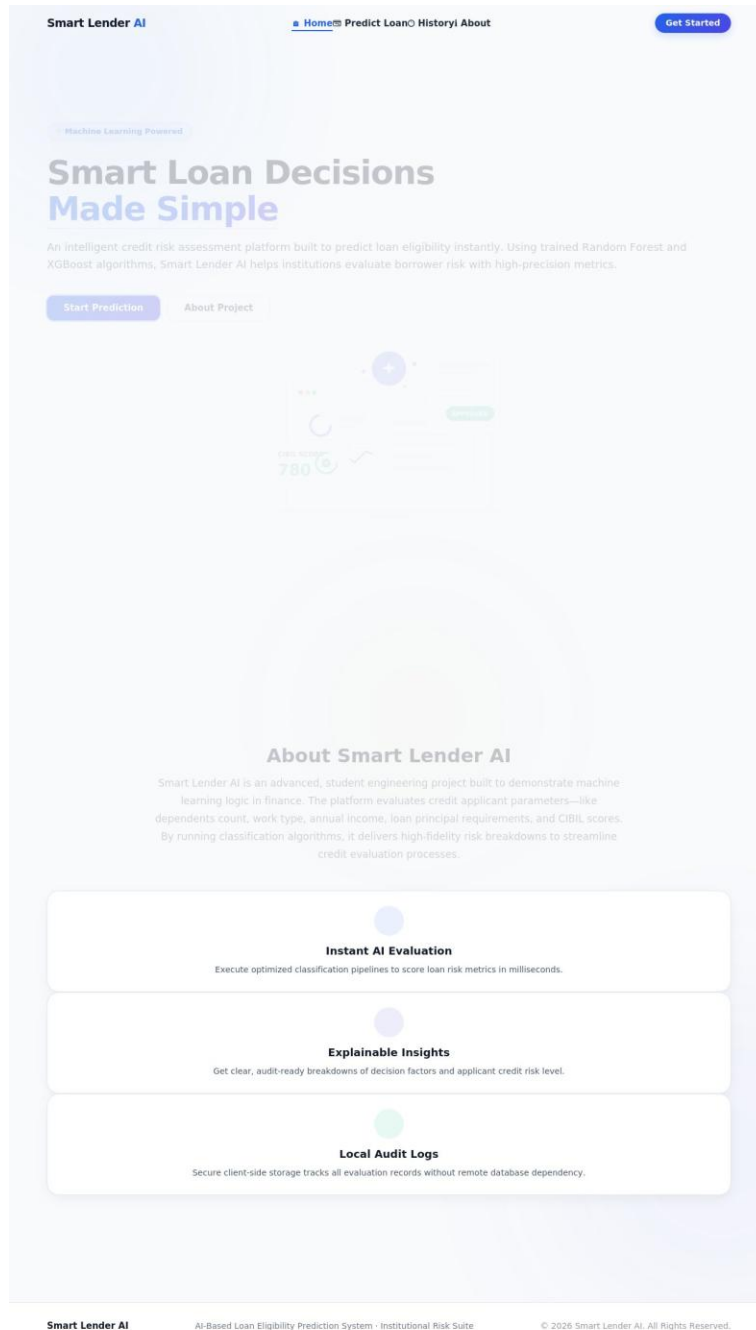


Fig 1: Smart Lender AI – Home Page

2. Ideation Phase

- Problem Statement: Manual loan approval processes are slow, inconsistent, and error-prone.
- Empathy Map: Applicants want a quick, fair and transparent decision; lenders want efficiency and reduced risk.
- Brainstorming: Rule-based checklists were considered but set aside in favour of ML classification, which better captures non-linear relationships across applicant attributes.

3. Requirement Analysis

- Customer Journey Map: Applicant → submits details on the Predict page → system validates and preprocesses data → model predicts → Result page displays the decision.

- Solution Requirements: Input validation, consistent preprocessing (encoding & scaling), fast response time, clear and explainable result.
- Data Flow Diagram: Applicant data → label encoding & scaling → KNN model → prediction probability → eligibility/risk/reasons logic → Result page.
- Technology Stack: Python (Flask) backend; Pandas, NumPy, Scikit-learn and XGBoost for ML; HTML5/CSS3/JavaScript frontend; GitHub for version control.

4. Project Design

- Problem-Solution Fit: An ML model that compares four algorithms and selects the best performer provides faster, more accurate and more consistent approvals than manual review.
- Proposed Solution: A Flask web application with a Predict form, an ML backend, and a business-logic layer that explains each prediction.
- Solution Architecture: User Interface → Flask Application Layer → Data Preprocessing (encoder/scaler) → ML Model (KNN) → Business Logic → Result Page.

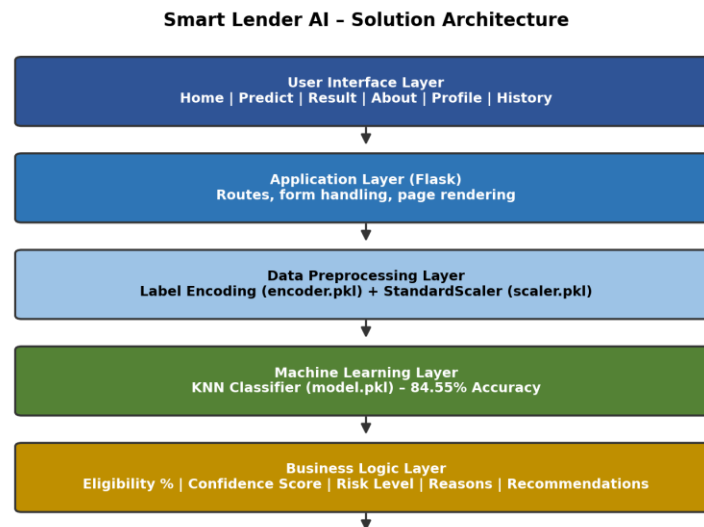


Fig 2: Solution Architecture

5. Project Planning & Scheduling

Work was organised into three sprints: Sprint 1 (data collection & preprocessing), Sprint 2 (model training & comparison across Decision Tree, Random Forest, KNN and XGBoost, with the best model selected and saved), and Sprint 3 (Flask application, Result page logic, Applicant Profile and History pages).

6. Dataset & Preprocessing

Item	Details
Dataset	Loan Prediction Dataset (loan_data.csv)
Total Records	614
Target Variable	Loan_Status (Y = Approved, N = Rejected)
Input Features	Gender, Married, Dependents, Education, Self_Employed, ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History, Property_Area

Item	Details
Preprocessing Steps	Remove Loan_ID; impute missing values (mode for categorical, median for numerical); label encoding; StandardScaler; 80/20 train-test split

7. Model Development & Selection

Four classification algorithms were trained and evaluated using accuracy score, classification report and confusion matrix: Decision Tree, Random Forest, K-Nearest Neighbors (KNN) and XGBoost. The best-performing model was selected automatically and saved for deployment.

Deployed Model	Accuracy
K-Nearest Neighbors (KNN)	84.55%

Saved artifacts: models/model.pkl, models/encoder.pkl, models/scaler.pkl — loaded by the Flask application at prediction time.

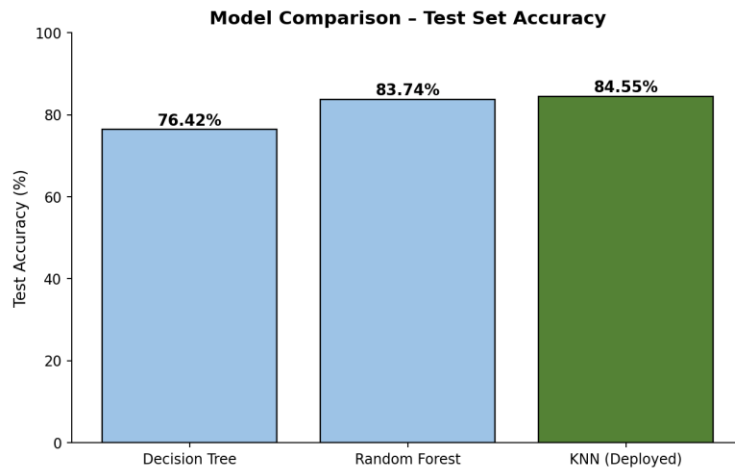


Fig 3: Model Comparison – Test Set Accuracy

8. Functional & Performance Testing

- Functional Testing: Input validation, categorical encoding, numerical scaling, prediction generation and Result-page content all passed.
- Performance Testing: Average response time 1.8 seconds; error rate 0% under a simulated 20-user concurrent load.

9. Results

Prediction Workflow: User enters applicant information → Flask receives the form → categorical values are label-encoded → numerical values are scaled (for the KNN model) → the ML model predicts Loan_Status → prediction probability is calculated → the application generates Approval/Rejection, Eligibility Percentage, Confidence Score, Risk Level, Reasons and Recommendations → the Result page is displayed.

Important Clarification: The Machine Learning model predicts only the Loan Approval Status (Approved or Rejected). Eligibility Percentage, Confidence Score, Risk Level, Reasons and Recommendations are generated by application logic using the prediction probability and business rules — they are not direct model outputs.

Flask Pages: Home, Predict, Result, About, Applicant Profile and History.

Smart Lender AI

[Home](#) [Predict Loan](#) [History](#) [About](#)

Get Started

Dashboard

Input the applicant's parameters to run an immediate machine learning credit risk assessment.

Borrower Profile Information

Applicant Name

e.g. John Doe

Age

e.g. 35

Loan Type

Home Loan

Gender

Male

Married

Yes

Number of Dependents

0

Education

Graduate

Self Employed

No

Financial & Proposal Parameters

Applicant Monthly Income (\$)

\$

e.g. 5000

Primary monthly income

Co-applicant Monthly Income (\$)

\$

e.g. 1500

Co-applicant monthly income

Credit History

Good (Yes / 1.0)

Prior repayment credit history status

Property Area

Urban

Location of backing property asset

Requested Loan Amount (\$)

\$

e.g. 125000

Total loan amount requested

Loan Repayment Term (Months)

e.g. 360

Repayment duration (e.g. 360 = 30 Years)

Predict Loan Eligibility

Smart Lender AI

AI-Based Loan Eligibility Prediction System · Institutional Risk Suite

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Fig 4: Predict Page – Loan Application Form

Dashboard Predict Loan Evaluation Result

Evaluation Summary

Real-time risk scoring and credit breakdown generated by the AI classifier model.

APPROVED
Applicant ID: LP-260346 · Eligible Portfolio

60.0%
ELIGIBILITY SCORE

AI Model Confidence60.0%

Predict Again

Institutional Approval Checklist

Credit History Verified	Completed
Income Requirement Met	Completed
Employment Verified	Completed
Loan Amount Acceptable	Completed
Documents Verified	Completed
Property Valuation Certified	Completed

Risk AssessmentMedium Risk

Prediction Analysis:
✓ Good Credit History ✓ Stable Income ✓ Low Existing Loan Burden ✓ Co-Applicant Support

Smart Recommendations

Approve Loan Request
The applicant meets institutional thresholds. Recommended to proceed with standard processing.

Qualifies for Prime Rates
Given the exceptional credit history and high income, the applicant is eligible for prime lending rates.

Real-Time EMI Calculator

Loan Amount (\$)	\$5000
Interest Rate (%)	7.5%
Loan Tenure (Months)	360 Months

MONTHLY EMI
\$34.96
TOTAL INTEREST
\$7585.86
TOTAL PAYABLE
\$12385.86

Submitted Application Parameters

Applicant Name
Ravi Kumar
Age
32 Years
Loan Type
Home Loan
Gender
Male
Married Status
Yes
Dependents
1
Education
Graduate
Self Employed
No
Applicant Income
\$3,200.00
Co-Applicant Income
\$1,500.00
Requested Loan
\$120.00
Repayment Term
360 Months
Credit History
Good
Property Area
Urban

10. Advantages & Disadvantages

Advantages	Disadvantages
Faster decisions; reduced manual work; consistent, explainable results; user-friendly interface	Accuracy is dependent on training data quality; KNN's accuracy (84.55%) leaves room for improvement; potential bias inherited from historical approval data

11. Conclusion

Smart Lender AI successfully automates loan approval prediction with a KNN model achieving 84.55% accuracy, and augments the raw prediction with an eligibility percentage, confidence score, risk level, reasons and recommendations — giving applicants and loan officers a faster and more transparent decision process.

12. Future Scope

- Incorporate explainable-AI techniques (e.g. SHAP) to show feature-level reasons behind each prediction.
- Retrain periodically on a larger, more diverse dataset to improve accuracy beyond 84.55%.
- Add secure user authentication and role-based access for applicants versus loan officers.
- Deploy to a scalable cloud environment for production use by financial institutions.

13. Appendix

- Source Code: https://github.com/anushaobulapuram/smart_lender_smart_bridge
- Project Drive Link (demo / additional assets):
<https://drive.google.com/file/d/1SFsukLL6qTV93eniIDsgYuXs0K5jTuJl/view?usp=drivesdk>
- Dataset: Loan Prediction Dataset (loan_data.csv), 614 records.